Events Explanation – OLAP Analyses

Task 5

Group 20

1. Overview

This document explains the patterns found in the three Saiku analyses with the historical events stored in the data warehouse (`BRIDGE_EVENT`, linked to `DIM_COUNTRY` and `DIM_TIME`).

Analysis	Cube	Measures	File
A. Long-run growth	Economy	GDP per capita, by year and decade	`analysis_1_growth.xlsx`
B. European comparison	Development	GDP, GDP per capita, GNI per capita, life expectancy	`analysis_2_comparison.xlsx`
C. Structural transformation	Sector	Value added (% of GDP)	`analysis_3_transformation.xlsx`

All measures use the `avg` aggregator, a decade shows the average of its years, and a country group shows the unweighted average of its countries.

2. Analysis A – Long-run Growth

Portugal is compared with the Western European countries that have complete series since 1900. The last column is Portugal as a percentage of the average of the five other countries.

Decade	Portugal	France	Germany	Italy	Spain	Greece	PT / avg.
1900s	1 997	4 666	4 921	3 534	2 776	2 396	55%
1910s	1 950	4 997	4 996	3 856	2 982	2 226	51%
1920s	2 298	6 305	5 479	4 314	3 631	3 264	50%
1930s	2 679	6 891	6 704	4 700	3 385	3 907	52%
1940s	2 977	5 662	7 031	4 296	3 420	2 530	65%
1950s	3 869	9 732	8 794	7 270	4 407	3 950	57%
1960s	6 236	14 367	14 244	12 199	7 165	6 896	57%
1970s	10 740	20 828	19 554	17 514	12 043	12 261	65%
1980s	13 892	25 022	23 980	22 593	15 235	14 777	68%
1990s	19 553	29 539	28 656	28 395	21 689	17 489	78%
2000s	24 400	35 167	36 611	34 529	30 340	25 533	75%
2010s	25 584	37 297	44 372	34 361	32 370	23 416	74%
2020s	27 277	37 741	45 934	34 479	32 504	23 594	78%

Turning points and events (Portugal)

Period	Pattern in the Analysis	Events That Explain It
1900–1925	Stagnation; the gap widens (55% → 50%)	State bankruptcy (1902), end of the monarchy (1910), World War I (1916–1917), political instability and inflation (1920), Banco de Portugal scandal (1924)
1926–1939	Slow recovery	Military coup (1926), fiscal stabilisation under Salazar (1928), Estado Novo (1933), economic reconstruction plan (1935)
1940s	Relative gain (65%) without real growth	Neutrality in World War II and wolfram exports (1941), while Germany (1945–1946) and Greece (1944–1945) collapse
1953–1973	Fastest growth (6 236 → 10 740 per decade)	Planos de Fomento (1953, 1959, 1967), founding member of EFTA (1960), industry surpasses agriculture (1960), free trade agreement with the EEC (1972); GDP per capita reaches 56% of the EC-12 average (1973)
1974–1985	Fall in 1975 and stagnation in 1983–1984	First oil crisis (1973), Carnation Revolution and end of the colonial war (1974), nationalisations (1975), IMF bailouts (1977 and 1983)
1986–2000	Strongest convergence (68% → 78%)	EEC accession and structural funds (1986), privatisations (1989–1990); GDP per capita passes the 1973 level (1991) and reaches 70% of the EU-12 (2000)
2001–2019	Stagnation; divergence from Spain (90% → 79%)	Stagnation from 2003, global financial crisis (2008), debt crisis (2010), Troika bailout (2011), unemployment peak (2013), BES collapse (2014)
2020	Sharp fall and rebound	COVID-19 pandemic; GDP contracts sharply and tourism collapses temporarily

3. Analysis B – European Comparison

Total GDP is included as requested, but it is not comparable between countries of different size (Portugal: 296 billion, Western European average: 1 629 billion in 2022), so the comparison uses the per-capita measures and life expectancy.

Portugal versus Western Europe, since 1975

	Portugal, % of W. Europe		Life Expectancy	
Year	GDP p.c.	GNI p.c.	Portugal	W. Europe
1975	64%	45%	68.3	72.5
1985	66%	34%	72.8	75.1
1990	75%	44%	74.0	76.2
2000	74%	56%	76.3	78.3
2008	68%	56%	78.5	80.5
2013	67%	55%	80.7	81.6
2022	72%	56%	81.6	81.9

- **Convergence after 1986.** GDP per capita rises from 66% to 75% of the Western European average in five years, after EEC accession (1986) and the end of the IMF programmes (1977, 1983). Spain joins the EEC in the same year and Greece in 1981.
- **Divergence after 2000.** The relative position falls back to 67% by 2013, with the adoption of the euro (1999), stagnation from 2003, the financial and debt crises (2008–2010) and the Troika bailout (2011). It

recovers to 72% only in 2022.

- **Life expectancy converges almost fully.** The gap falls from 4.1 years (1975) to 0.3 years (2022), health converged even when income did not.

Portugal versus Eastern Europe, since 1990

	GDP per Capita			Life Expectancy		
Year	**Portugal**	**C./E. Europe**	**Post-Soviet**	**Portugal**	**C./E. Europe**	**Post-Soviet**
1990	17 256	9 555	15 621	74.0	70.3	70.0
1995	19 262	10 027	9 963	75.3	71.1	67.6
2004	24 143	15 748	17 057	77.7	73.7	72.0
2008	25 591	20 481	21 884	78.5	74.7	72.7
2019	28 056	27 977	27 783	81.7	77.3	76.8
2022	28 992	30 091	29 384	81.6	76.8	75.9

- **Transition shock in the early 1990s.** The fall of communism (Poland, Hungary, Czechia and Romania, 1989), the Baltic independence movements (1988–1991) and the market transition (Balcerowicz Plan, 1990) cause a deep recession: Post-Soviet GDP per capita falls from 15 621 to 9 963 between 1990 and 1995.
- **Fast catch-up after 2000.** Both Eastern groups grow much faster than Portugal, especially after EU accession (2004; Romania 2007), and overtake Portugal in GDP per capita around 2020, while Portugal stagnates.
- **Portugal keeps a clear lead in life expectancy**, about 5 years above both Eastern groups in 2022.

4. Analysis C – Structural Transformation

Sector value added is available for Portugal from 1995, so the 1990s decade only includes 1995–1999. The table shows the drill-down from Broad Sector to Sector (value added, % of GDP).

Broad Sector	**Sector**	**1990s**	**2000s**	**2010s**	**2020s**
Primary	Agriculture	4.0	2.5	1.9	2.1
Secondary	Industry	25.2	22.1	19.3	18.8
Secondary	Manufacturing	16.1	13.1	11.9	12.0
Tertiary	Services	58.5	62.7	65.9	66.0

- **Deindustrialisation.** Industry loses 6.4 points of GDP and manufacturing 4.1 points. The events show the end of heavy industry: Lisnave Margueira closes (2000) and the Siderurgia Nacional blast furnace is shut down (2001). The new automotive industry (Autoeuropa, from 1995) partly offsets the loss and keeps manufacturing at about 12% since 2010.
- **Growth of services.** Services rise from 58.5% to 66%, driven by EU structural funds and infrastructure (motorways from 1991, Expo 98 and the Vasco da Gama bridge in 1998, Lisbon Metro), the euro (1999), privatisations (Siderurgia Nacional 1991, EDP 1995) and the telecoms reform (1992).
- **Agriculture halves** (4.0% → 2.1%), the earlier shift from agriculture to industry happened before the data starts (industry surpassed agriculture in 1960).

Comparison by country group, 2010s

Sector	**Portugal**	**W. Europe**	**C./E. Europe**	**Post-Soviet**	**EU**
Agriculture	1.9	1.7	3.3	3.3	1.6
Industry	19.3	21.3	30.0	23.6	22.3
Manufacturing	11.9	14.6	19.7	14.0	14.8
Services	65.9	66.8	54.7	61.2	65.3

Portugal has a Western European structure, with a smaller manufacturing base than the Western and EU

averages. Central/Eastern Europe keeps a much larger industry (30% of GDP), linked to the foreign investment that followed the market transition of the 1990s.

5. Reading Notes and Limitations

- **Unweighted averages.** Country-group values are simple averages of their countries small and large countries have the same weight.
- **Decade averages depend on the available years.** Decades with gaps (Hungary, Poland and Romania before 1920 and in World War II) or partial decades (sector data from 1995, the 2020s) average fewer years.
- **The Secondary level is not meaningful.** Manufacturing is part of industry, so the Secondary value (the average of both) does not represent the secondary sector the comparison is made at the Sector level. Value-added shares also do not add up to 100% (net taxes).
- **Different units.** GDP per capita is in 2011 international dollars (Maddison, until 2022) GNI per capita is in current US dollars (WDI), so it also reflects exchange rates, as in the escudo depreciation of the early 1980s.
- **Neither correlation nor timing is causation.** The events are a chronology, the links in this document are interpretations of the timing, not measured effects.